AYMAN MAHDIA KHAN

Chittagong, Bangladesh
aymanmahdia3@gmail.com LinkedIn ResearchGate ORCID Website

RESEARCH INTEREST

Hydrological Modeling, Flood Risk Management, Surface Water Management, Stormwater Management, Climate Change Adaptation, Groundwater Modelling

EDUCATION

B.Sc. in Civil Engineering, Chittagong University of Engineering & Technology (CUET) Feb 2020 – Jun 2025

- CGPA 3.21/4.00 (162.5 credits); merit position 36th of 130 (top 28%).
- Departmental **Merit Scholarship** for six semesters.

RESEARCH EXPERIENCE

Research Assistant, Dept. of Civil Engineering, CUET Nov 2025 – Present

"Integrated Assessment of Groundwater, Surface Water, and Rainwater Availability under Climate Change Scenarios in the Chattogram Hill Tracts (Integrated Water CHT)" | Project Leader: Prof. Dr. Aysha Akter | DRE-CUET funded (BDT 1.5M)

- Modelling water availability under RCP 4.5 and 8.5 across 13,295 km² of three hill districts, using **SWAT+ with BWDB rainfall, SRTM DEM and Sentinel-1/2 data**.
- Mapped surface-water (Rangamati) and groundwater (Bandarban) potential zones using frequency-ratio analysis in ArcGIS; two conference manuscripts submitted **(ICACE 2026, ICERIE 2027)**.
- **Led 14 field visits**: surveyed 33 springs, wells, and gauges; ran 26 household surveys, 3 focus groups, and 10 meetings with local and district government offices to turn model results into water-supply options.

Undergraduate Thesis (Grade: A+), Supervisor: Prof. Dr. Aysha Akter, CUET Apr 2024 – Jun 2025

"A Case Study on the Stormwater Drainage System Analysis and Design under Extreme Rainfall Events", Agrabad, Chittagong

- Built and calibrated an urban drainage model of the Agrabad catchment in **EPA SWMM 5.2** (94 sub-catchments, 245 conduits, 186 junctions) against observed flows and simulated a recent observed storm and a 50-year design storm.
- Redesigned **87 key conduits** that **reduced flooded junctions by 29.6%** and **peak flood volume by 39.9%** under both the recent and 50-year storm scenarios in Bangladesh's one of the most flood-exposed commercial districts.

PEER-REVIEWED JOURNAL PUBLICATION

- Akter, A., **Khan, A. M.**, & Farooq, S. M. (2026). *Surface water–groundwater–rainwater interactions in the Chittagong Hill Tracts, Bangladesh: A critical review of modeling and decision-support approaches*. *Hydrology, 13*(9), Article 242. https://doi.org/10.3390/hydrology13090242

CONFERENCE PAPERS

- Sultana, S., Shafique, S., Akter, S. T., Roy, C., **Khan, A. M.**, & Zishan, M. B. (2026, September 26). *Seismic risk assessment in Chattogram: An explainable machine learning approach*. iCONEECT 2026, Premier University, Chattogram, Bangladesh.
- Mahin, M. M. R., Hossain, H., **Khan, A. M.**, Alvi, M. M. I., & Banu, M. M. (2025, December). *A sustainable approach to develop eco-concrete using fly ash and silica fumes as SCMs*. ICERIE 2025, RUET, Rajshahi, Bangladesh.
- Srabony, S. C., Khan, M. S., **Khan, A. M.**, & Hameem, M. M. (2024, December). *Influence of industrial wastes and nano materials on the properties of recycled aggregate concrete: A review*. ICACE 2024, CUET, Chattogram, Bangladesh.

TECHNICAL SKILLS

Hydrological & hydraulic modelling: EPA SWMM 5.2, SWAT+ (research); HEC-HMS, SewerGEMS (training)
GIS & remote sensing: ArcGIS (ArcMap 10.8), QGIS; SRTM DEM, Landsat, Sentinel-1/2

Programming & data analysis: Python (Pandas, GeoPandas, scikit-learn), machine learning (Random Forest, Gradient Boosting, XGBoost), Google Colab, MS Excel

Project & sustainability tools: MS Project, MS Visio, OpenLCA, Life-365

PROFESSIONAL TRAINING & FIELD EXPERIENCE

Industrial Training (credit-bearing, Grade A) | Institute of Water Modelling (IWM), Dhaka (3 weeks) Dec 2024

- Built 2D river morphology models in MIKE 21 (DHI), with data preparation and mapping in ArcMap 10.8.
- Trained in urban flood forecasting and water resources management in IWM's technical division.
- Visited the Goran Chatbari stormwater pumping station and Dasherkandi Sewage Treatment Plant.

Software-Aided Civil Engineering Design & Analysis | EDGE Project, BCC, ICT Division (80 hours) Jun 2024 – Oct 2024

- Government-funded training in GIS, HEC-HMS, SWMM, SewerGEMS, PLAXIS, ETABS and PMS.

Short courses & field visits: Hydrographic survey with Chittagong Port Authority (Feb 2025, 1 day); Sanitation for Urban Low-Income Communities, CUET & ITN-BUET, including a visit to Halishahar STP (Aug 2025, 2 days).

COMMUNITY ENGAGEMENT & SCIENCE COMMUNICATION

Flood Rehabilitation Project | ACI Student Chapter, CUET Oct 2024 – Jan 2025

- Led the construction of a home for a flood-displaced family in Sundarpur, Fatikchhari: managed site assessment, estimation, and costing on a BDT 100,000 budget, and coordinated approvals with the administration and local authorities.

Exhibitor, Bangladesh Innovation Fair 2026 | University Zone, CUET Civil Engineering Sep 2026

- Presented the Integrated Water CHT project with an interactive demonstration of groundwater, surface-water, and rainwater potential mapping, engaging over 700 visitors across three days.

AWARDS & COMPETITIONS

Top 20 international finalist (11th overall; 2nd from South Asia), American Concrete Institute (ACI) Spring 2022 Concrete Solutions Competition.

- Designed a flood-resistant modular house for flood-prone areas using Autoclaved Aerated Concrete (AAC) and Rice Husk Ash (RHA); the design first won the Intra-CUET Concrete Solutions Competition (ACI Student Chapter, 2022).

Champion, National Case Study Competition, Civil Explosion CUET, 2023.

- Designed an academic building for a forested site with minimal environmental impact.

Finalist, Face the Case 2.0, IEEE CUET Student Branch, 2022.

- Solved a real-world case addressing corruption and black ticketing in the railway system.

LEADERSHIP & SERVICE

General Secretary (May 2024 – May 2025) and Assistant General Secretary (Mar 2023 – May 2024), ACI Student Chapter, CUET; led a chapter of over 300 members; organised technical events, competitions, and the flood rehabilitation project.

Joint Organising Secretary, Green for Peace (Jul 2024 – Jun 2025), environmental advocacy, community outreach.

Judge, Envision 2.0, ASCE Student Chapter, CUET (Dec 2025); **Speaker**, Case Study Session, ASCE Student Chapter (Oct 2024); **Volunteer**, 7th ICACE (Dec 2024).

LANGUAGE PROFICIENCY

IELTS Academic (May 2026): Overall 7.0 (Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0); **Bangla (Native)**

REFERENCES

Dr. Aysha Akter
Professor, Dept. of Civil Engineering, CUET
Email: aysha_akter@cuet.ac.bd

Dr. G. M. Sadiqul Islam
Professor, Dept. of Civil Engineering, CUET
Email: gmsislam@cuet.ac.bd